10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 1.3 — Modulus and Conjugate

Unit 1 | Solved Exercise | New Book (PECTAA)

Exercise 1.3 (Solved)

1. Find the modulus of the following complex numbers:

(i) $4 + 3i$ (ii) $-5 - 4i$ (iii) $\frac{3}{5} - \frac{4}{5}i$

(iv) $-\sqrt{2} - \sqrt{3}\,i$

For $z = x + iy$, modulus $|z| = \sqrt{(x^2 + y^2)}$. Modulus is always a positive real number.

(i) $4 + 3i$

$|z| = \sqrt{(4)^2 + (3)^2}$

$= \sqrt{16 + 9}$

$= \sqrt{25}$

Answer: 5

(ii) $-5 - 4i$

$|z| = \sqrt{(-5)^2 + (-4)^2}$

$= \sqrt{25 + 16}$

$= \sqrt{41}$

Answer: $\sqrt{41}$

(iii) $3/5 - 4/5\,i$

$|z| = \sqrt{(3/5)^2 + (-4/5)^2}$

$= \sqrt{9/25 + 16/25}$

$= \sqrt{25/25}$

$= \sqrt{1}$

Answer: 1

(iv) $-\sqrt{2} - \sqrt{3}\,i$

$|z| = \sqrt{(-\sqrt{2})^2 + (-\sqrt{3})^2}$

$= \sqrt{2 + 3}$

$= \sqrt{5}$

Answer: $\sqrt{5}$

2. If $z_1 = 2 + 7i$ and $z_2 = 4 - 3i$, then verify that:

(i) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

(ii) $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$

(iii) $\overline{(z_1/z_2)} = \overline{z_1} \,/\, \overline{z_2}$

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A bar over an expression means its conjugate, e.g. $\bar{z}$ = conjugate of z. To take a conjugate, change the sign of the imaginary part.

(i) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

L.H.S. $= \overline{z_1 + z_2}$

$= \overline{2 + 7i + 4 - 3i}$

$= \overline{6 + 4i}$

$= 6 - 4i$

R.H.S. $= \overline{z_1} + \overline{z_2}$

$= (2 - 7i) + (4 + 3i)$

$= 6 - 4i$

Answer: L.H.S. = R.H.S., hence verified

(ii) $\overline{z_1 z_2} = \overline{z_1} \cdot \overline{z_2}$

$z_1 z_2 = (2 + 7i)(4 - 3i)$

$= 8 - 6i + 28i - 21i^2$

$= 8 + 22i + 21$

$= 29 + 22i$

L.H.S. $= \overline{29 + 22i}$

$= 29 - 22i$

R.H.S. $= (2 - 7i)(4 + 3i)$

$= 8 + 6i - 28i - 21i^2$

$= 8 - 22i + 21$

$= 29 - 22i$

Answer: L.H.S. = R.H.S., hence verified

(iii) $\overline{(z_1/z_2)} = \overline{z_1} / \overline{z_2}$

$\frac{z_1}{z_2} = \frac{2 + 7i}{4 - 3i} \times \frac{4 + 3i}{4 + 3i}$

$= \frac{8 + 6i + 28i + 21i^2}{16 + 9}$

$= \frac{8 + 34i - 21}{25}$

$= \frac{-13 + 34i}{25}$

L.H.S. $= \overline{z_1/z_2}$

$= -13/25 - 34i/25$

R.H.S. $= \overline{z_1} / \overline{z_2}$

$= \frac{2 - 7i}{4 + 3i} \times \frac{4 - 3i}{4 - 3i}$

$= \frac{8 - 6i - 28i + 21i^2}{16 + 9}$

$= \frac{8 - 34i - 21}{25}$

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$= \frac{-13 - 34i}{25}$

Answer: L.H.S. = R.H.S., hence verified

3. If z = 5 − 2i, then verify that:

(i) conjugate of the conjugate = z

(ii) $|z| = |\bar{z}|$

(iii) $|z| = |-z|$

(iv) $z \cdot \bar{z} = |z|^2$

(v) $|z| = |\overline{-z}|$

(i) Conjugate of the conjugate

$z = 5 - 2i$

$\bar{z} = 5 + 2i$

Taking the conjugate again: $\overline{5 + 2i} = 5 - 2i$

$= z$

Answer: conjugate of the conjugate = z, hence verified

(ii) $|z| = |\bar{z}|$

$|z| = \sqrt{(5)^2 + (-2)^2}$

$= \sqrt{29}$

$\bar{z} = 5 + 2i$

$|\bar{z}| = \sqrt{(5)^2 + (2)^2}$

$= \sqrt{29}$

Answer: $|z| = |\bar{z}|$, hence verified

(iii) $|z| = |-z|$

$-z = -(5 - 2i)$

$= -5 + 2i$

$|-z| = \sqrt{(-5)^2 + (2)^2}$

$= \sqrt{25 + 4}$

$= \sqrt{29}$

Answer: $|z| = |-z|$, hence verified

(iv) $z \cdot \bar{z} = |z|^2$

$z \cdot \bar{z} = (5 - 2i)(5 + 2i)$

$= (5)^2 - (2i)^2$

$= 25 - 4i^2$

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$= 25 + 4$

$= 29$

$|z|^2 = (\sqrt{29})^2$

$= 29$

Answer: $\mathbf{z \cdot \bar{z} = |z|^2}$**, hence verified**

(v) $\mathbf{|z| = |\overline{-z}|}$

$\overline{-z} = -(5 + 2i)$

$= -5 - 2i$

$|\overline{-z}| = \sqrt{(-5)^2 + (-2)^2}$

$= \sqrt{25 + 4}$

$= \sqrt{29}$

Answer: $\mathbf{|z| = |\overline{-z}|}$**, hence verified**

4. If $z = 4 - 3i$, then verify that $|z| = |-z| = |\bar{z}| = |-\bar{z}|$.

$|z| = \sqrt{(4)^2 + (-3)^2}$

$= \sqrt{16 + 9}$

$= \sqrt{25}$

$= 5$

$-z = -4 + 3i$

$|-z| = \sqrt{(-4)^2 + (3)^2}$

$= \sqrt{25}$

$= 5$

$\bar{z} = 4 + 3i$

$|\bar{z}| = \sqrt{(4)^2 + (3)^2}$

$= \sqrt{25}$

$= 5$

$\overline{-z} = -4 - 3i$

$|\overline{-z}| = \sqrt{(-4)^2 + (-3)^2}$

$= \sqrt{25}$

$= 5$

Answer: $\mathbf{|z| = |-z| = |\bar{z}| = |\overline{-z}| = 5}$**, hence verified**

5. If $z_1 = 2 + 3i$ and $z_2 = -1 + i$, then evaluate: (i) $\mathrm{Re}(z_1 z_2)$ (ii) $\mathrm{Im}(z_1 z_2)$

$z_1 z_2 = (2 + 3i)(-1 + i)$

$= -2 + 2i - 3i + 3i^2$

$= -2 - i + 3(-1) \quad \because i^2 = -1$

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$= -5 - i$

Comparing with $a + bi$: $a = -5$, $b = -1$

Answer: (i) $\mathrm{Re}(z_1 z_2) = -5$ (ii) $\mathrm{Im}(z_1 z_2) = -1$

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